# Foundations of Mean squared error

[Mean squared error]

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

### 1.0 Content Block

a = ( n − 1 ) γ 2 + n 2 + n n = n + 1 + n − 1 n γ 2 . {\displaystyle a={\frac {(n-1)\gamma _{2}+n^{2}+n}{n}}=n+1+{\frac {n-1}{n}}\gamma _{2}.}

### 2.0 Content Block

MSE ⁡ ( S n − 1 2 ) = 1 n ( μ 4 − n − 3 n − 1 σ 4 ) = 1 n ( γ 2 + 2 n n − 1 ) σ 4 , {\displaystyle \operatorname {MSE} (S_{n-1}^{2})={\frac {1}{n}}\left(\mu _{4}-{\frac {n-3}{n-1}}\sigma ^{4}\right)={\frac {1}{n}}\left(\gamma _{2}+{\frac {2n}{n-1}}\right)\sigma ^{4},}

### 3.0 Content Block

where σ 2 {\displaystyle \sigma ^{2}} is the population variance. For a Gaussian distribution this is the best unbiased estimator of the population mean, that is the one with the lowest MSE (and hence variance) among all unbiased estimators. One can check that the MSE above equals the inverse of the Fisher information (see Cramér–Rao bound). But the same sample mean is not the best estimator of the population mean, say, for a uniform distribution.

### 4.0 Content Block

In other words, the MSE is the mean ( 1 n ∑ i = 1 n ) {\textstyle \left({\frac {1}{n}}\sum _{i=1}^{n}\right)} of the squares of the errors ( Y i − Y i ^ ) 2 {\textstyle \left(Y_{i}-{\hat {Y_{i}}}\right)^{2}} . This is an easily computable quantity for a particular sample (and hence is sample-dependent). In matrix notation,

### 5.0 Content Block

S a 2 = n − 1 a S n − 1 2 = 1 a ∑ i = 1 n ( X i − X ¯ ) 2 {\displaystyle S_{a}^{2}={\frac {n-1}{a}}S_{n-1}^{2}={\frac {1}{a}}\sum _{i=1}^{n}\left(X_{i}-{\overline {X}}\,\right)^{2}}

### 6.0 Content Block

MSE ⁡ ( S a 2 ) = E ⁡ [ ( n − 1 a S n − 1 2 − σ 2 ) 2 ] = E ⁡ [ ( n − 1 ) 2 a 2 S n − 1 4 − 2 ( n − 1 a S n − 1 2 ) σ 2 + σ 4 ] = ( n − 1 ) 2 a 2 E ⁡ [ S n − 1 4 ] − 2 ( n − 1 a ) E ⁡ [ S n − 1 2 ] σ 2 + σ 4 = ( n − 1 ) 2 a 2 E ⁡ [ S n − 1 4 ] − 2 ( n − 1 a ) σ 4 + σ 4 E ⁡ [ S n − 1 2 ] = σ 2 = ( n − 1 ) 2 a 2 ( γ 2 n + n + 1 n − 1 ) σ 4 − 2 ( n − 1 a ) σ 4 + σ 4 E ⁡ [ S n − 1 4 ] = MSE ⁡ ( S n − 1 2 ) + σ 4 = n − 1 n a 2 ( ( n − 1 ) γ 2 + n 2 + n ) σ 4 − 2 ( n − 1 a ) σ 4 + σ 4 {\displaystyle {\begin{aligned}\operatorname {MSE} (S_{a}^{2})&=\operatorname {E} \left[\left({\frac {n-1}{a}}S_{n-1}^{2}-\sigma ^{2}\right)^{2}\right]\\&=\operatorname {E} \left[{\frac {(n-1)^{2}}{a^{2}}}S_{n-1}^{4}-2\left({\frac {n-1}{a}}S_{n-1}^{2}\right)\sigma ^{2}+\sigma ^{4}\right]\\&={\frac {(n-1)^{2}}{a^{2}}}\operatorname {E} \left[S_{n-1}^{4}\right]-2\left({\frac {n-1}{a}}\right)\operatorname {E} \left[S_{n-1}^{2}\right]\sigma ^{2}+\sigma ^{4}\\&={\frac {(n-1)^{2}}{a^{2}}}\operatorname {E} \left[S_{n-1}^{4}\right]-2\left({\frac {n-1}{a}}\right)\sigma ^{4}+\sigma ^{4}&&\operatorname {E} \left[S_{n-1}^{2}\right]=\sigma ^{2}\\&={\frac {(n-1)^{2}}{a^{2}}}\left({\frac {\gamma _{2}}{n}}+{\frac {n+1}{n-1}}\right)\sigma ^{4}-2\left({\frac {n-1}{a}}\right)\sigma ^{4}+\sigma ^{4}&&\operatorname {E} \left[S_{n-1}^{4}\right]=\operatorname {MSE} (S_{n-1}^{2})+\sigma ^{4}\\&={\frac

{n-1}{na^{2}}}\left((n-1)\gamma _{2}+n^{2}+n\right)\sigma ^{4}-2\left({\frac {n-1}{a}}\right)\sigma ^{4}+\sigma ^{4}\end{aligned}}}

### 7.0 Content Block

Interpretation An MSE of zero, meaning that the estimator θ ^ {\displaystyle {\hat {\theta }}} predicts observations of the parameter θ {\displaystyle \theta } with perfect accuracy, is ideal (but typically not possible). Values of MSE may be used for comparative purposes. Two or more statistical models may be compared using their MSEs—as a measure of how well they explain a given set of observations: An unbiased estimator (estimated from a statistical model) with the smallest variance among all unbiased estimators is the best unbiased estimator or MVUE (Minimum-Variance Unbiased Estimator). Both analysis of variance and linear regression techniques estimate the MSE as part of the analysis and use the estimated MSE to determine the statistical significance of the factors or predictors under study. The goal of experimental design is to construct experiments in such a way that when the observations are analyzed, the MSE is close to zero relative to the magnitude of at least one of the estimated treatment effects. In one-way analysis of variance, MSE can be calculated by the division of the sum of squared errors and the degree of freedom. Also, the f-value is the ratio of the mean squared treatment and the MSE. MSE is also used in several stepwise regression techniques as part of the determination as to how many predictors from a candidate set to include in a model for a given set of observations.

### 8.0 Content Block

An even shorter proof can be achieved using the well-known formula that for a random variable X {\textstyle X} , E ( X 2 ) = Var ■ ( X ) + ( E ( X ) ) 2 {\textstyle \mathbb {E} (X^{2})=\operatorname {Var} (X)+(\mathbb {E} (X))^{2}} . By substituting X {\textstyle X} with, θ ^ – θ {\textstyle {\hat {\theta }}-\theta } , we have

### 9.0 Content Block

MSE ■ ( θ ^ ) = E [ ( θ ^ – θ ) 2 ] = Var ■ ( θ ^ – θ ) + ( E [ θ ^ – θ ] ) 2 = Var ■ ( θ ^ ) + Bias 2 ■ ( θ ^ , θ ) {\displaystyle {\begin{aligned}\operatorname {MSE} ({\hat {\theta }})&=\mathbb {E} [({\hat {\theta }}-\theta )^{2}]\\&=\operatorname {Var} ({\hat {\theta }}-\theta )+(\mathbb {E} [{\hat {\theta }}-\theta ])^{2}\\&=\operatorname {Var} ({\hat {\theta }})+\operatorname {Bias} ^{2}({\hat {\theta }},\theta )\end{aligned}}}

### 10.0 Content Block

Loss function Squared error loss is one of the most widely used loss functions in statistics, though its widespread use stems more from mathematical convenience than considerations of actual loss in applications. Carl Friedrich Gauss, who introduced the use of mean squared error, was aware of its arbitrariness and was in agreement with objections to it on these grounds. The mathematical benefits of mean squared error are particularly evident in its use at analyzing the performance of linear regression, as it allows one to partition the variation in a dataset into variation explained by the model and variation explained by randomness.